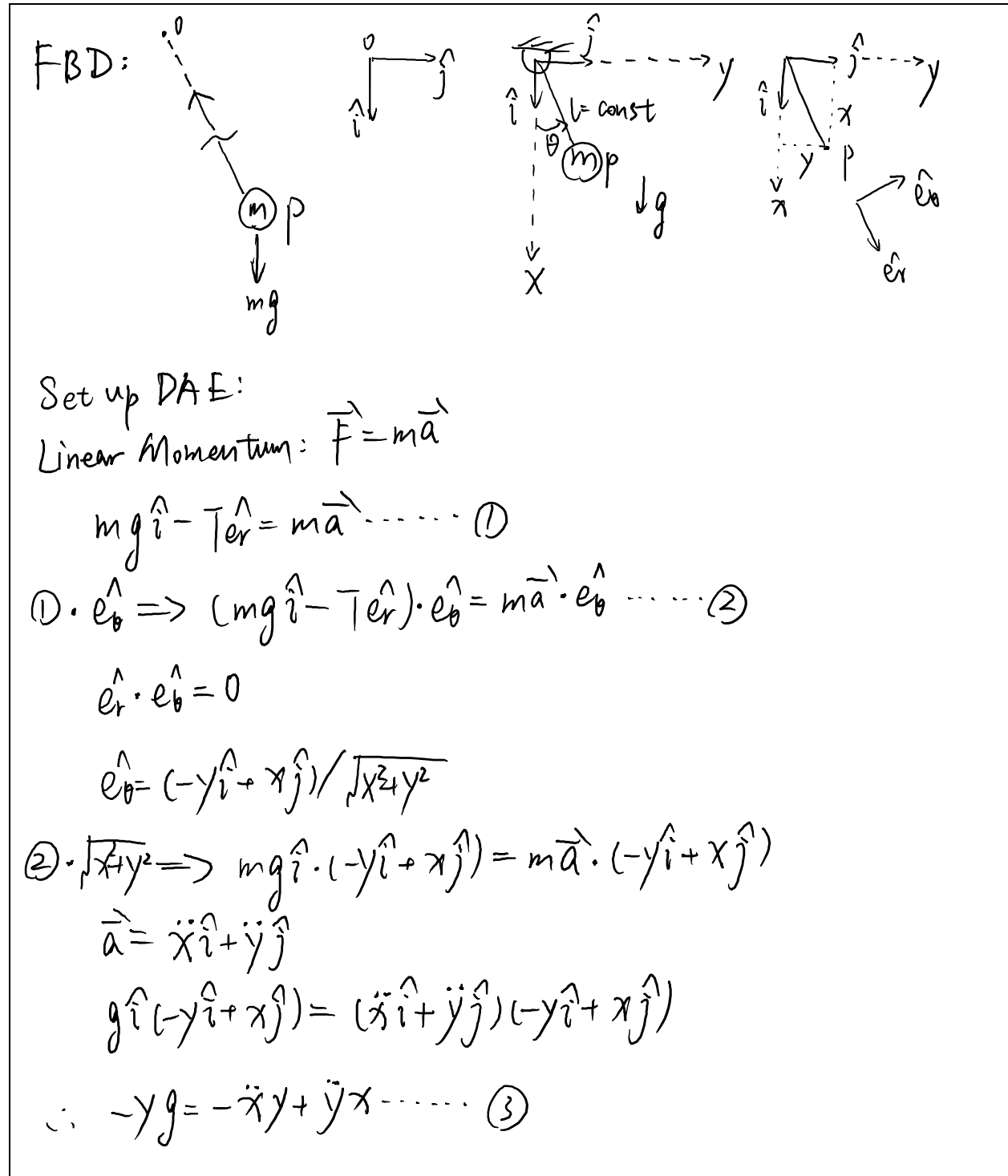


20. Pendulum with an awkward parameterization.

Problem Restatement: Write Equations of Motion using y for minimal coordinate.

Write Equations of Motion of the pendulum in Cartesian Coordinates:



Set up DAE:

Linear Momentum: $\vec{F} = m\vec{a}$

$$mg\hat{i} - T\hat{e}_r = m\vec{a} \dots \dots (1)$$

$$\textcircled{1} \cdot \hat{e}_\theta \Rightarrow (mg\hat{i} - T\hat{e}_r) \cdot \hat{e}_\theta = m\vec{a} \cdot \hat{e}_\theta \dots \dots (2)$$

$$\hat{e}_r \cdot \hat{e}_\theta = 0$$

$$\hat{e}_\theta = (-y\hat{i} + x\hat{j}) / \sqrt{x^2 + y^2}$$

$$\textcircled{2} \cdot \sqrt{x^2 + y^2} \Rightarrow mg\hat{i} \cdot (-y\hat{i} + x\hat{j}) = m\vec{a} \cdot (-y\hat{i} + x\hat{j})$$

$$\vec{a} = \ddot{x}\hat{i} + \ddot{y}\hat{j}$$

$$g\hat{i} \cdot (-y\hat{i} + x\hat{j}) = (\ddot{x}\hat{i} + \ddot{y}\hat{j}) \cdot (-y\hat{i} + x\hat{j})$$

$$\therefore -yg = -\ddot{x}y + \ddot{y}x \dots \dots (3)$$

$$\therefore x^2 + y^2 = l^2 = \text{const}$$

$$\therefore \frac{d}{dt}(x^2 + y^2) = 0 \quad \dot{x}x + \dot{y}y = 0 \quad \dot{x} = -\frac{y}{x}\dot{y} \dots \dots (4)$$

$$\therefore \frac{d^2}{dt^2}(x^2 + y^2) = 0 \quad \dot{x}^2 + x\ddot{x} + \dot{y}^2 + y\ddot{y} = 0 \dots \dots (5)$$

$$\therefore x = \sqrt{l^2 - y^2} \dots \dots (6)$$

$$(5) \Rightarrow \ddot{x} = \frac{1}{x}(-\dot{x}^2 - \dot{y}^2 - y\ddot{y})$$

$$\begin{aligned} (4), (6) \rightarrow (5) \Rightarrow \ddot{x} &= \frac{1}{x} \left(-\frac{y^2}{x^2} \cdot \dot{y}^2 - \dot{y}^2 - y\ddot{y} \right) \\ &= \frac{1}{\sqrt{l^2 - y^2}} \left[-\left(\frac{y^2}{l^2 - y^2} + \frac{l^2 - y^2}{l^2 - y^2} \right) \dot{y}^2 - y\ddot{y} \right] \\ &= \frac{1}{\sqrt{l^2 - y^2}} \left(\frac{l^2}{y^2 - l^2} \dot{y}^2 - y\ddot{y} \right) \dots \dots (7) \end{aligned}$$

$$\begin{aligned} (6), (7) \rightarrow (2) \Rightarrow -y\dot{g} &= -\dot{x}y + \ddot{x}x \\ -y\dot{g} &= -\frac{1}{\sqrt{l^2 - y^2}} \left(\frac{l^2}{y^2 - l^2} \cdot \dot{y}^2 - y\ddot{y} \right) y + \ddot{y}\sqrt{l^2 - y^2} \dots \dots (8) \end{aligned}$$

$$(8) \cdot \sqrt{l^2 - y^2} \Rightarrow -y\sqrt{l^2 - y^2}\dot{g} = -\frac{yl^2}{y^2 - l^2} \cdot \dot{y}^2 + y^2\ddot{y} + \ddot{y}(l^2 - y^2)$$

$$\ddot{y}l^2 = \frac{yl^2}{y^2 - l^2} \cdot \dot{y}^2 - y\sqrt{l^2 - y^2}\dot{g} \quad \left| \ddot{y} = \frac{y}{y^2 - l^2} \dot{y}^2 - \frac{y\sqrt{l^2 - y^2}}{l^2} \dot{g} \right|$$

$$\ddot{y} = \frac{y}{y^2 - l^2} \dot{y}^2 - \frac{y\sqrt{l^2 - y^2}}{l^2} \dot{g}$$

Use MATLAB Solve these DAE equations with numerical integration and initial conditions and parameters of choosing:

$$g=10 \text{ } l=5 \text{ } y_0=3 \text{ } dy_0=0$$

Quantitatively compare solution with a solution of the simple pendulum equations:

$$ddtheta = -g/l * \sin(theta)$$

myRHS

```
function derivative = myRHS(time,variable,myCoefficient)

g=myCoefficient.g;
l=myCoefficient.l;

y=variable(1);
vy=variable(2);
ay=y/(y*y-l*l)*vy*vy-y*sqrt(l*l-y*y)/l/l*g;

derivative=[vy;ay];
```

myRHS_theta

```
function derivative = myRHS_theta(time,variable,myCoefficient)

g=myCoefficient.g;
l=myCoefficient.l;

theta=variable(1);
dtheta=variable(2);
ddtheta=-g/l*sin(theta);

derivative=[dtheta;ddtheta];
```

Problem_20

```
clear all;

myCoefficient.g=10;
myCoefficient.l=5;
timeTotal=10;

opts.RelTol = 1e-6;
opts.AbsTol = 1e-6;
```

```

[timeArray,resultArray]=ode45(@Zhidi_Yang_RHS,[0 timeTotal],[3,0],opts,myCoefficient);
x=-resultArray(:,1);
y=-sqrt(myCoefficient.l*myCoefficient.l-x.^2);

[timeArray_2,resultArray_2]=ode45(@Zhidi_Yang_RHS_theta,[0
timeTotal],[atan(3/4),0],opts,myCoefficient);
theta=resultArray_2(:,1);

xx=-sin(theta)*myCoefficient.l;
yy=-cos(theta)*myCoefficient.l;

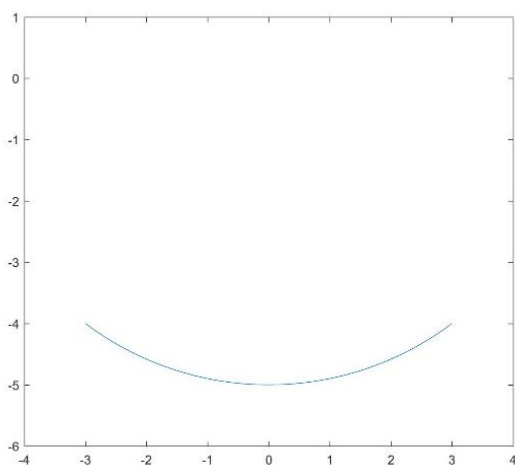
yEnd=atan(x(end)/y(end))
thetaEnd=theta(end)

subplot(1,2,2)
plot(xx,yy)
axis equal
axis([-4,4,-6,1]);

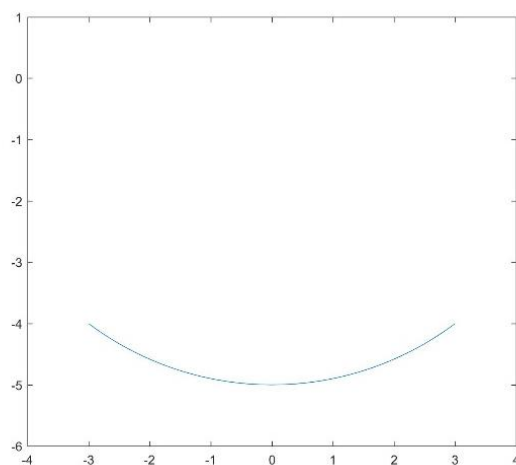
subplot(1,2,1)
plot(x,y)
axis equal
axis([-4,4,-6,1]);

```

When timeTotal=10:



Use y as minimal coordinate



Use θ as minimal coordinate

Accuracy:

Change tolerance and total time and verify the difference between using y as minimal coordinate and using θ as minimal coordinate. Consider the error as accuracy.

Problem_20

```

clear all;

myCoefficient.g=10;
myCoefficient.l=5;
timeTotal=10;

opts.RelTol = 1e-8;
opts.AbsTol = 1e-8;

accuracyArray=[];

for i=[1:1:10]
    timeTotal=timeTotal*2
    opts.RelTol = 1e-8;
    opts.AbsTol = 1e-8;
    for j=[1:1:5]
        opts.RelTol = opts.RelTol*10;
        opts.AbsTol = opts.AbsTol*10;
        [timeArray,resultArray]=ode45(@myRHS,[0 timeTotal],[3,0],opts,myCoefficient);
        x=-resultArray(:,1);
        y=-sqrt(myCoefficient.l*myCoefficient.l-x.^2);

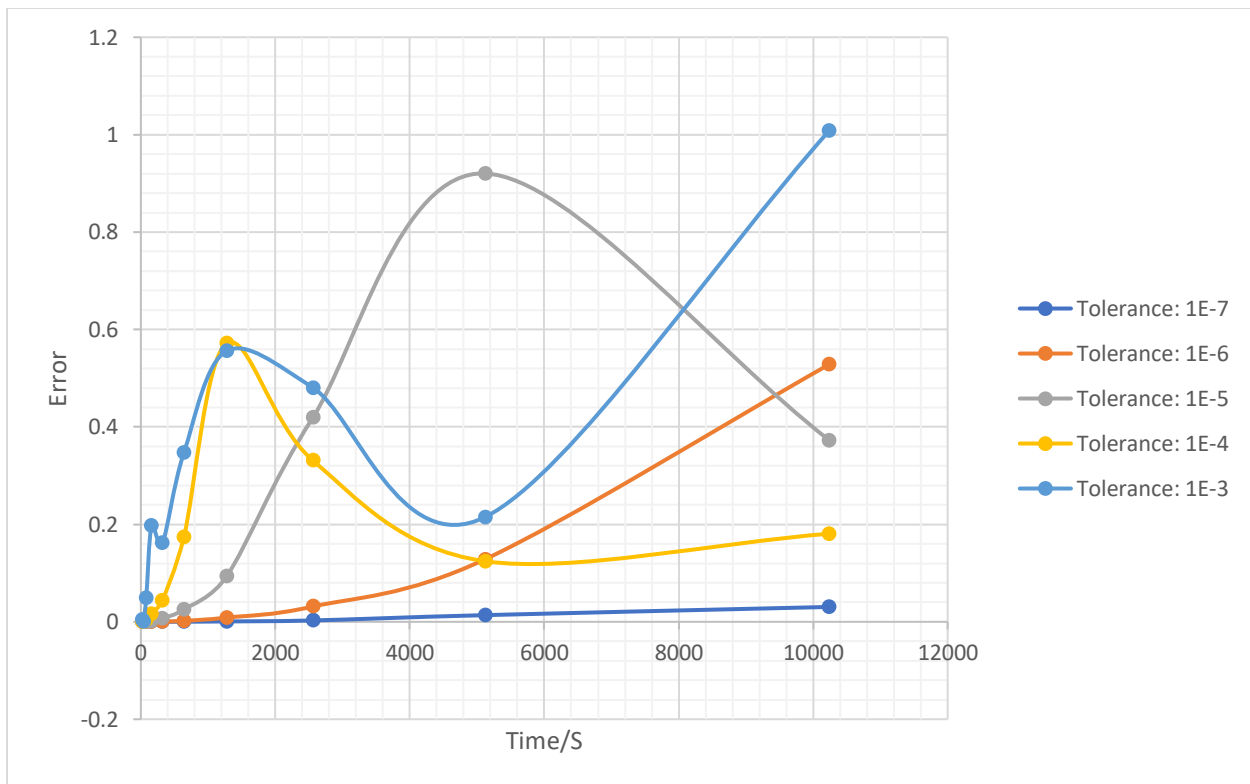
        [timeArray_2,resultArray_2]=ode45(@myRHS_theta,[0
timeTotal],[atan(3/4),0],opts,myCoefficient);
        theta=resultArray_2(:,1);

        xx=-sin(theta)*myCoefficient.l;
        yy=-cos(theta)*myCoefficient.l;

        % yEnd=atan(x(end)/y(end))
        % thetaEnd=theta(end)
        yEnd=y(end);
        thetaEnd=-cos(theta(end))*myCoefficient.l;
        accuracyArray(i,j)=yEnd-thetaEnd
    end
end
end

```

Time/S	Tolerance: 1E-7	Tolerance: 1E-6	Tolerance: 1E-5	Tolerance: 1E-4	Tolerance: 1E-3
20	2.60E-08	-6.40E-07	-2.85E-05	-0.000655465	0.004503557
40	-2.65E-07	-3.46E-06	-5.02E-05	-0.000582047	-0.001130778
80	3.93E-06	4.40E-05	0.000557462	0.00433765	0.048948062
160	1.56E-05	0.000173327	0.0021968	0.016675108	0.197224955
320	5.27E-05	0.000576189	0.007136468	0.044382721	0.161530967
640	-0.000186836	-0.002021492	-0.025436178	-0.173896743	0.346774128
1280	0.00081634	0.008659918	0.094183723	-0.571817683	-0.556387593
2560	-0.002793368	-0.031423718	-0.419544863	0.331011787	0.480577946
5120	0.013700776	0.1284198	-0.920364811	0.124354222	-0.215072915
10240	-0.030500898	-0.528544124	-0.372604823	0.180913666	1.008392541



As time goes to 10240 seconds, the error between using y as minimal coordinate and using θ as minimal coordinate are getting larger.

When tolerance is large than $1E-5$ and time equals to 10240 seconds, the error between using y as minimal coordinate and using θ as minimal coordinate are larger than 1 period.

As tolerance goes to zero, errors between using y as minimal coordinate and using θ as minimal coordinate are getting smaller.

When tolerance is $1E-7$, the result is accurate enough.